Vault-Door-5

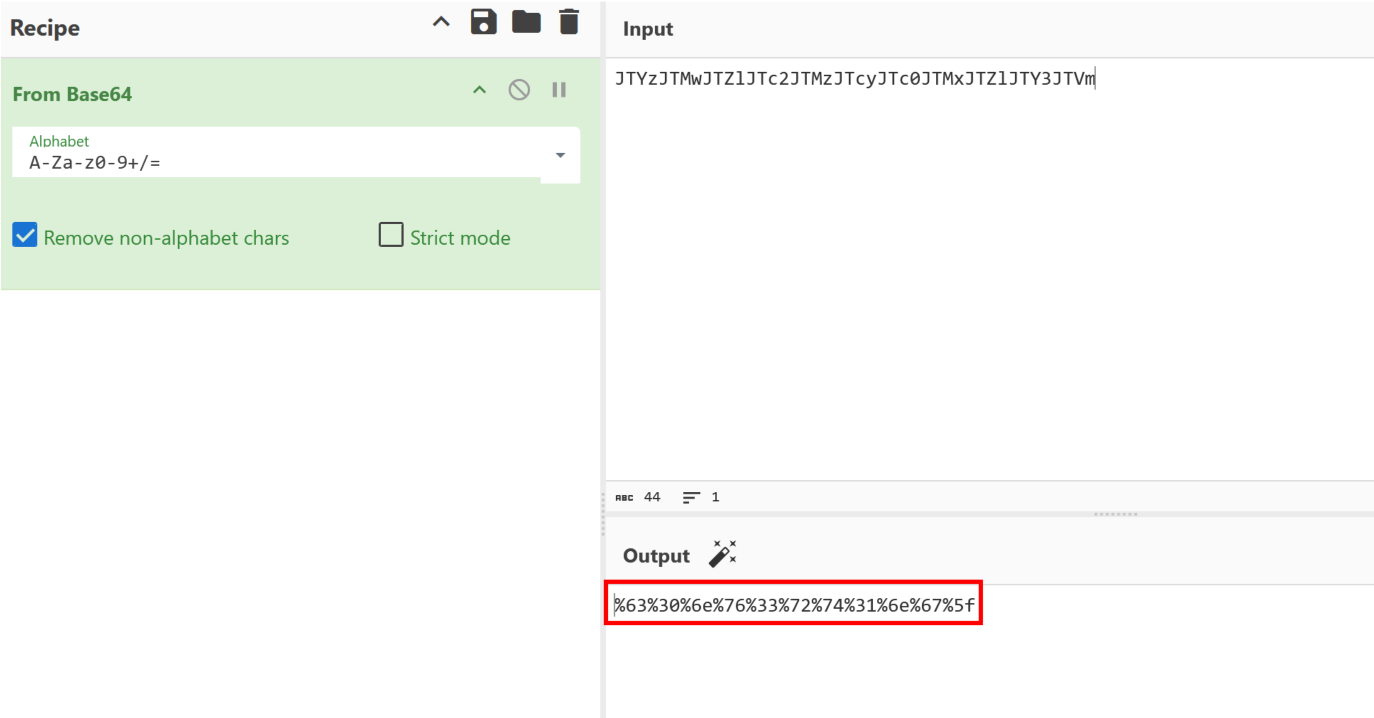
picoCTF › Reverse Engineering · + Tags

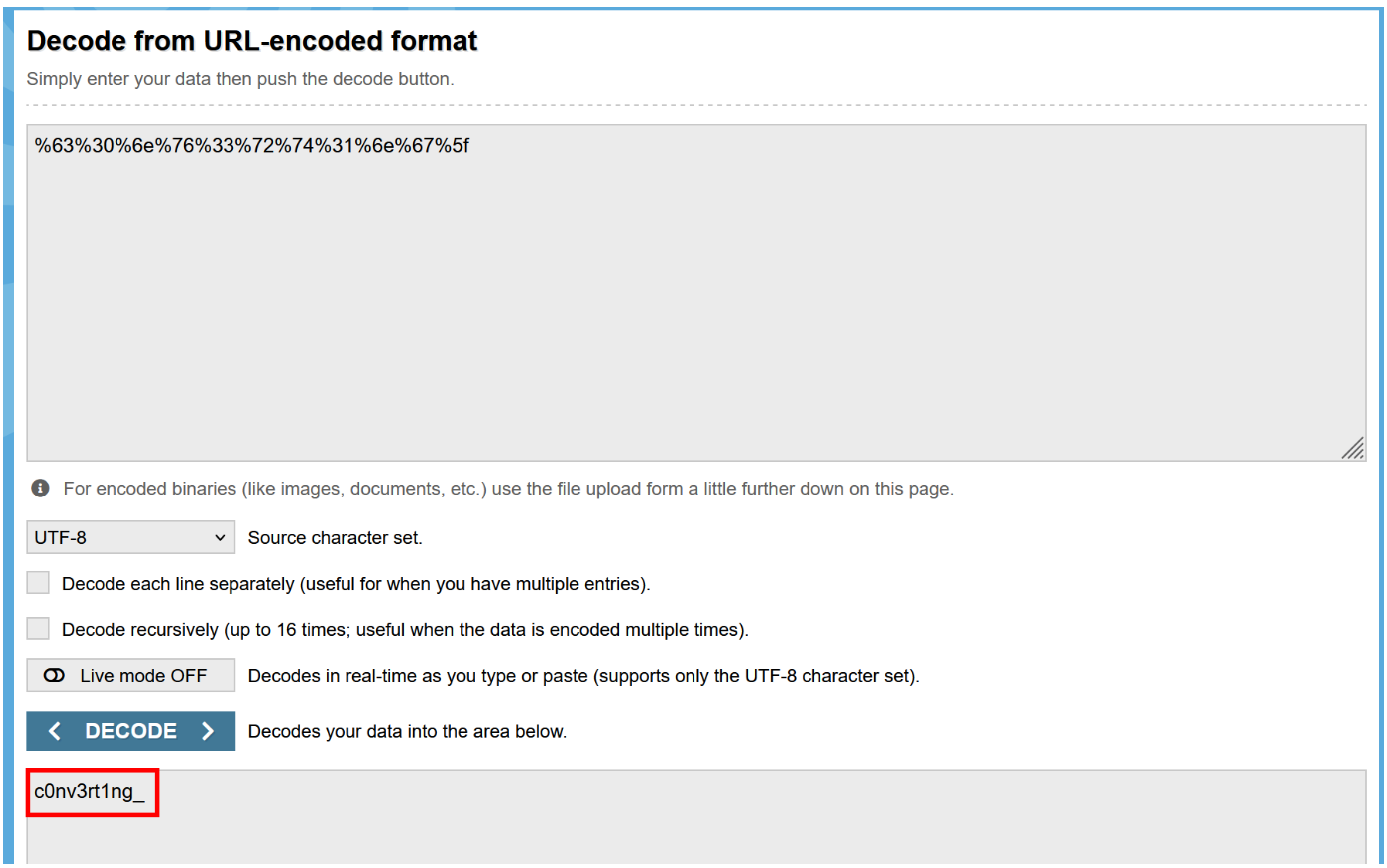
The code give is this:

```java
    //
    // -Minion #2414
    public String base64Encode(byte[] input) {
        return Base64.getEncoder().encodeToString(input);
    }

    // URL encoding is meant for web pages, so any double agent spies who steal
    // our source code will think this is a web site or something, defintely not
    // vault door! Oh wait, should I have not said that in a source code
    // comment?
    //
    // -Minion #2415
    public String urlEncode(byte[] input) {
        StringBuffer buf = new StringBuffer();
        for (int i=0; i<input.length; i++) {
            buf.append(String.format("%%%2x", input[i]));
        }
        return buf.toString();
    }

    public boolean checkPassword(String password) {
        String urlEncoded = urlEncode(password.getBytes());
        String base64Encoded = base64Encode(urlEncoded.getBytes());
        String expected = "JTYzJTMwJTZlJTc2JTMzJTcyJTc0JTMxJTZlJTY3JTVm"
                        + "JTY2JTcyJTMwJTZkJTVmJTYyJTYxJTM1JTY1JTVmJTM2"
                        + "JTM0JTVmJTY1JTY0JTMwJTYyJTM0JTMyJTM4JTM4";
        return base64Encoded.equals(expected);
    }
```

Here, it is visible that the password is first encrypted using url encodign and then, encrypted using base64 encoding. To reveal the flag, decode base64 first and url decode the given result at last.

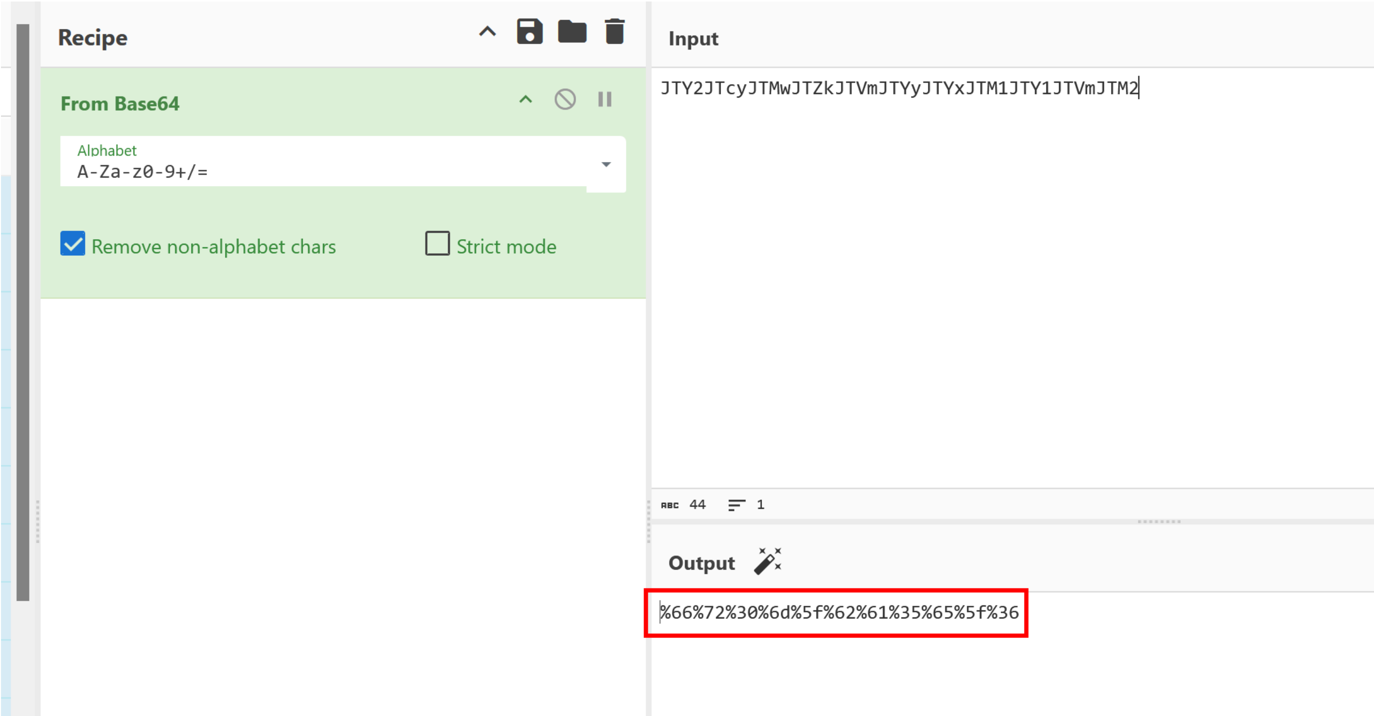
First Decryption

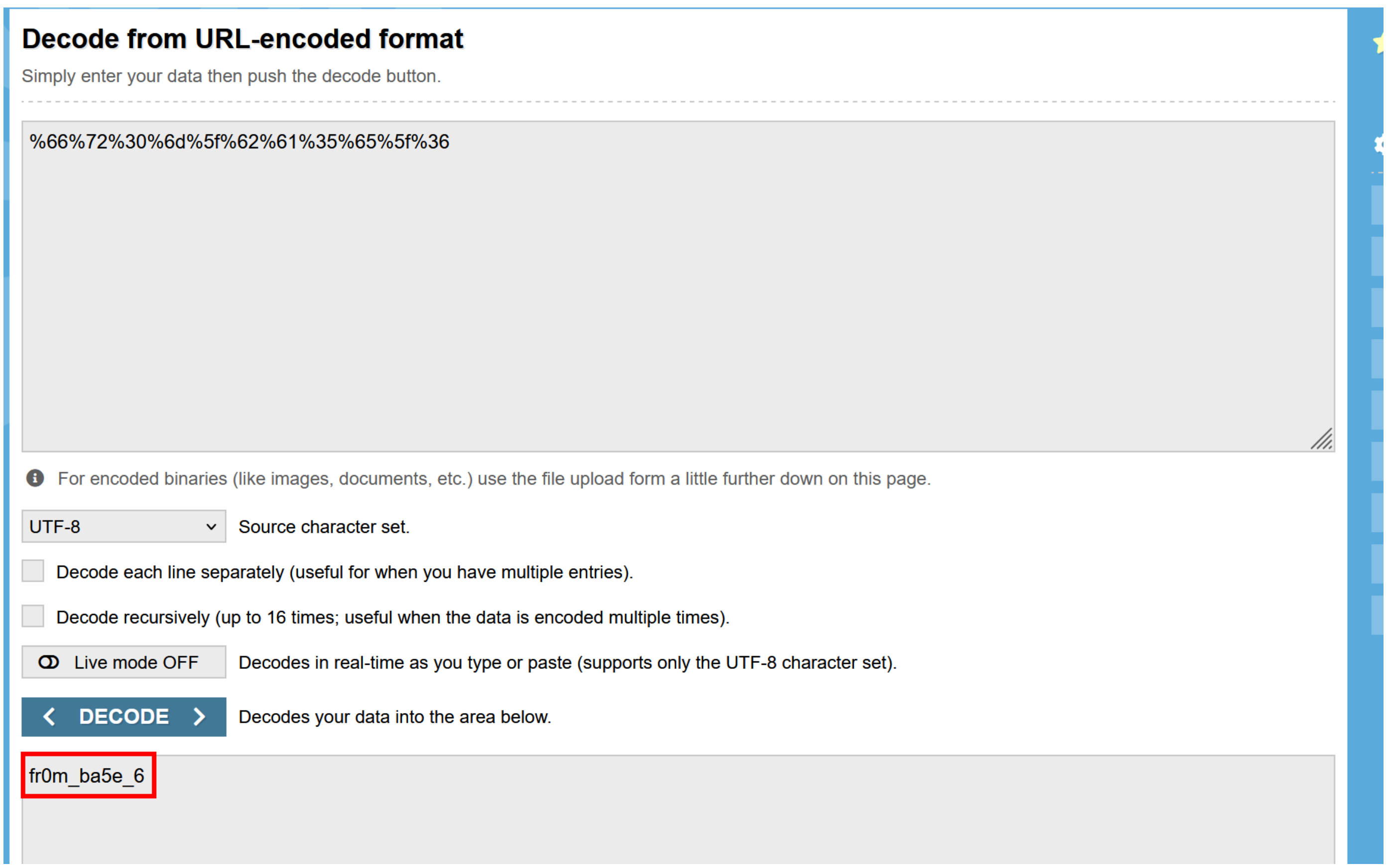




First part of the flag: `c0nv3rt1ng_`

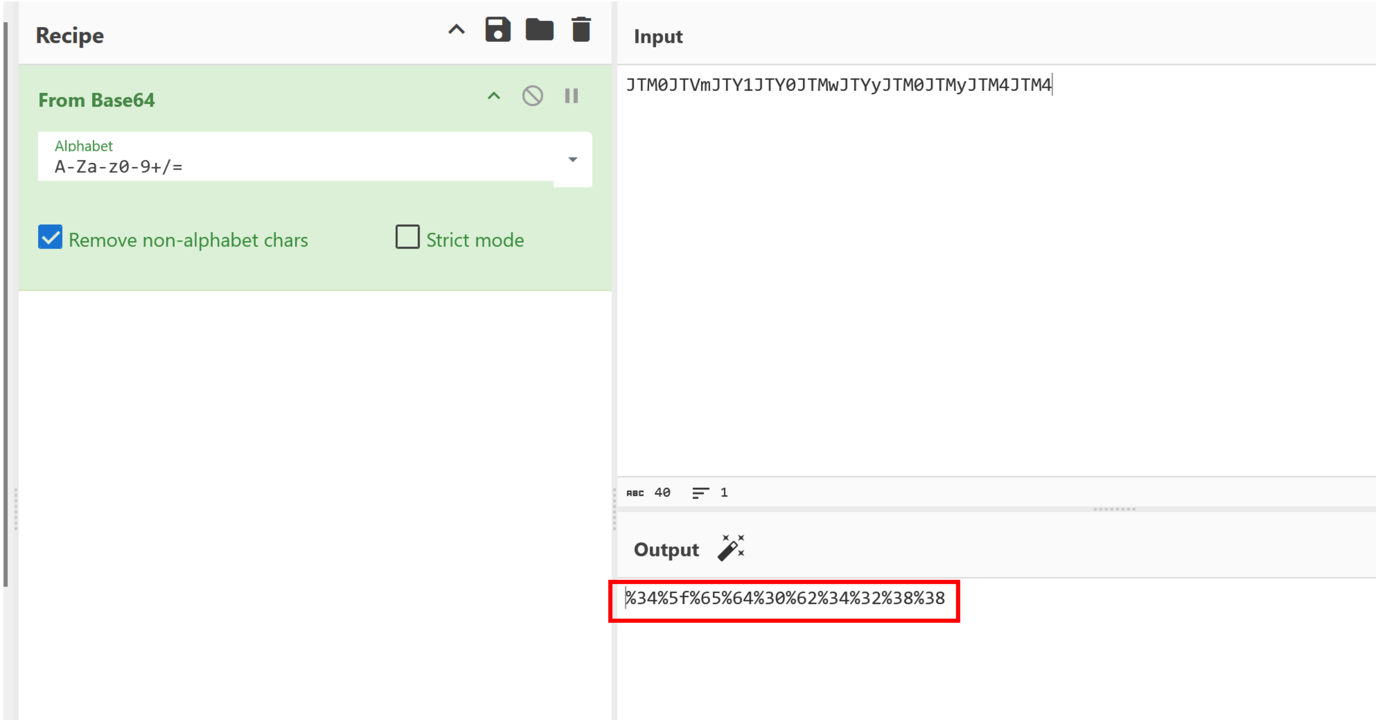
Second Decryption

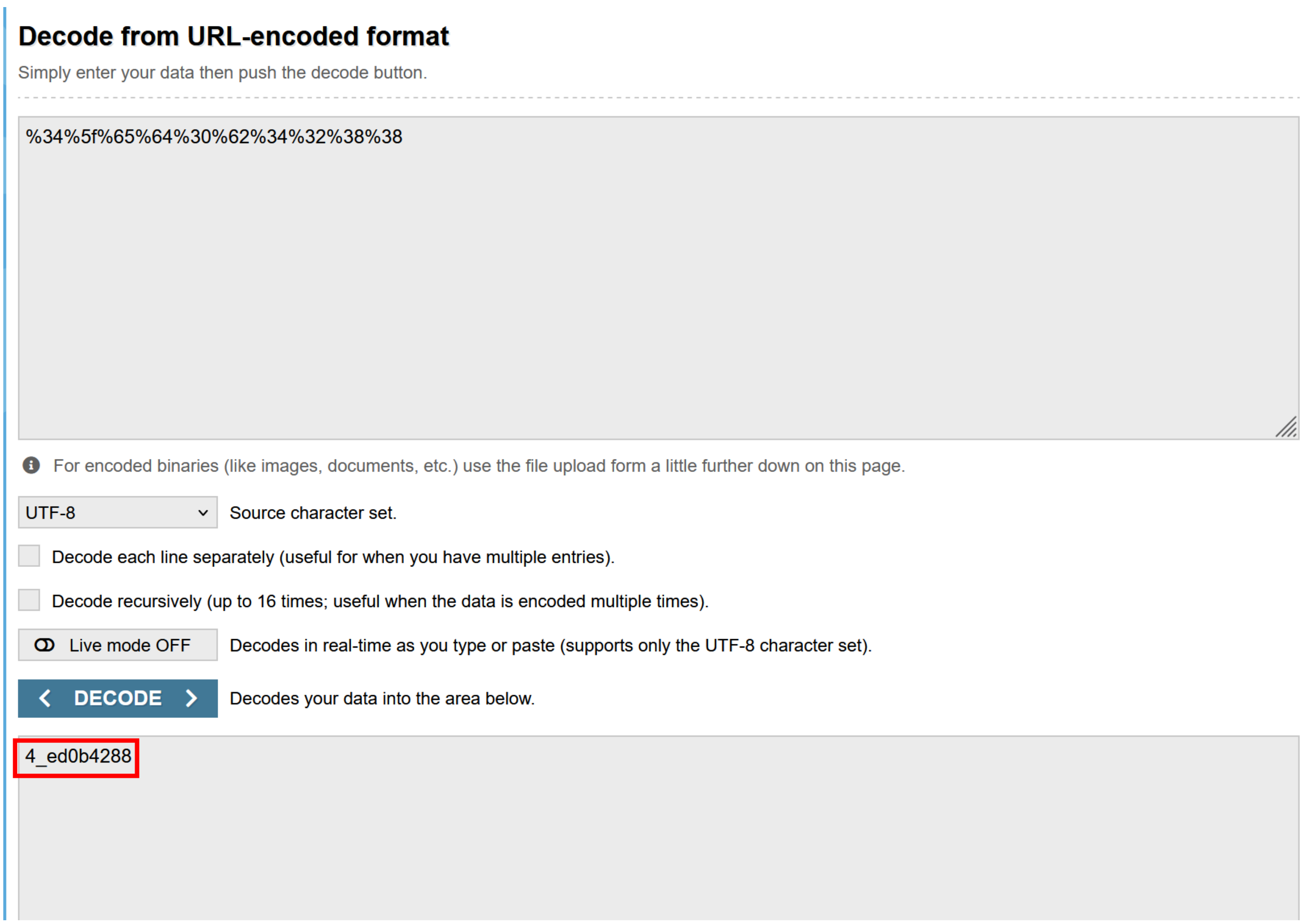




Second part of the flag: `fr0m_ba5e_6`

Third Decryption





Third part of the flag: `4_ed0b4288`

The flag: `picoCTF{c0nv3rt1ng_fr0m_ba5e_64_ed0b4288}`